

Combining overlap correction and back-to-back regression methods to analyse eye-tracking-EEG co-registration data during natural reading.

<Msc> Thesis Intro Talk - <Yanhong Xu>

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Supervisors: *Benedikt Ehinger, Jevri Hanna*

B2B main paper:

[Back-to-back regression: Disentangling the influence of correlated factors from multivariate observations](#)

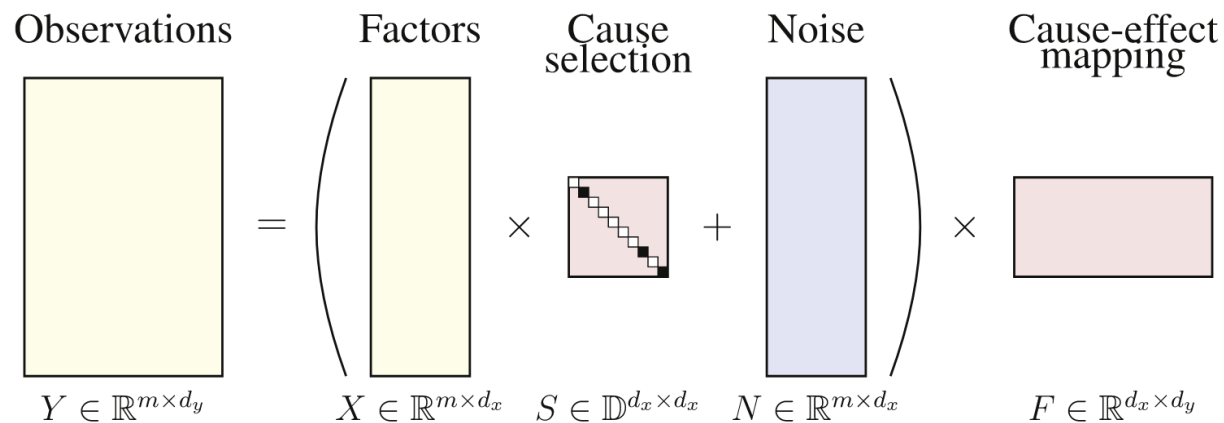
How can we identify which correlated factors independently explain high-dimensional brain responses?

- A reading task: successive words are separated.
- Correlated predictors: word function, word length, word frequency.
- Causal identification: Which feature independently explains brain responses?
 - word length: earlier effects in visual processing.
 - word frequency: later evoked N400 responses.

B2B Algorithm:

$$(X, Y) \rightarrow (X_{train}, Y_{train}) (X_{test}, Y_{test})$$

$$(X_{train}, Y_{train}) \rightarrow (X_1, Y_1) (X_2, Y_2)$$



$$X \sim P(X) \quad \hat{S} = \text{diag}((X_2^\top X_2 + \Lambda_X)^{-1} X_2^\top \underbrace{Y_2 (Y_1^\top Y_1 + \Lambda_Y)^{-1} Y_1^\top X_1}_{\substack{1) \hat{X}: \text{regression from } Y \text{ to } X \quad \triangleright \quad \hat{G} = (Y_1^\top Y_1 + \Lambda_Y)^{-1} Y_1^\top X_1 \\ 2) \text{regression from } X \text{ to } \hat{X} \quad \triangleright \quad \hat{H} = (X_2^\top X_2 + \Lambda_X)^{-1} X_2^\top Y_2 \hat{G}}})$$

$$N \sim P(N)$$

B2B feature importance evaluation:

$$(X, Y) \rightarrow (X_{train}, Y_{train}), (X_{test}, Y_{test})$$

- B2B is fit on the training data, and feature importance is evaluated on held-out test data by feature knockout.
- For each feature i , a full-model score and a knockout score are computed after removing feature i .
- A larger ΔR_i means that the feature provides more important information.

$$R_{full} = \text{corr}(X_{test}H, Y_{test}G)$$

$$\Delta R_i = R_{full} - R_{knockout(i)}$$

Algorithm 2: B2B feature importance.

Input: $X_{train} \in \mathbb{R}^{m \times d_x}$, $X_{test} \in \mathbb{R}^{m' \times d_x}$, $Y_{train} \in \mathbb{R}^{m \times d_y}$, $Y_{test} \in \mathbb{R}^{m' \times d_y}$,

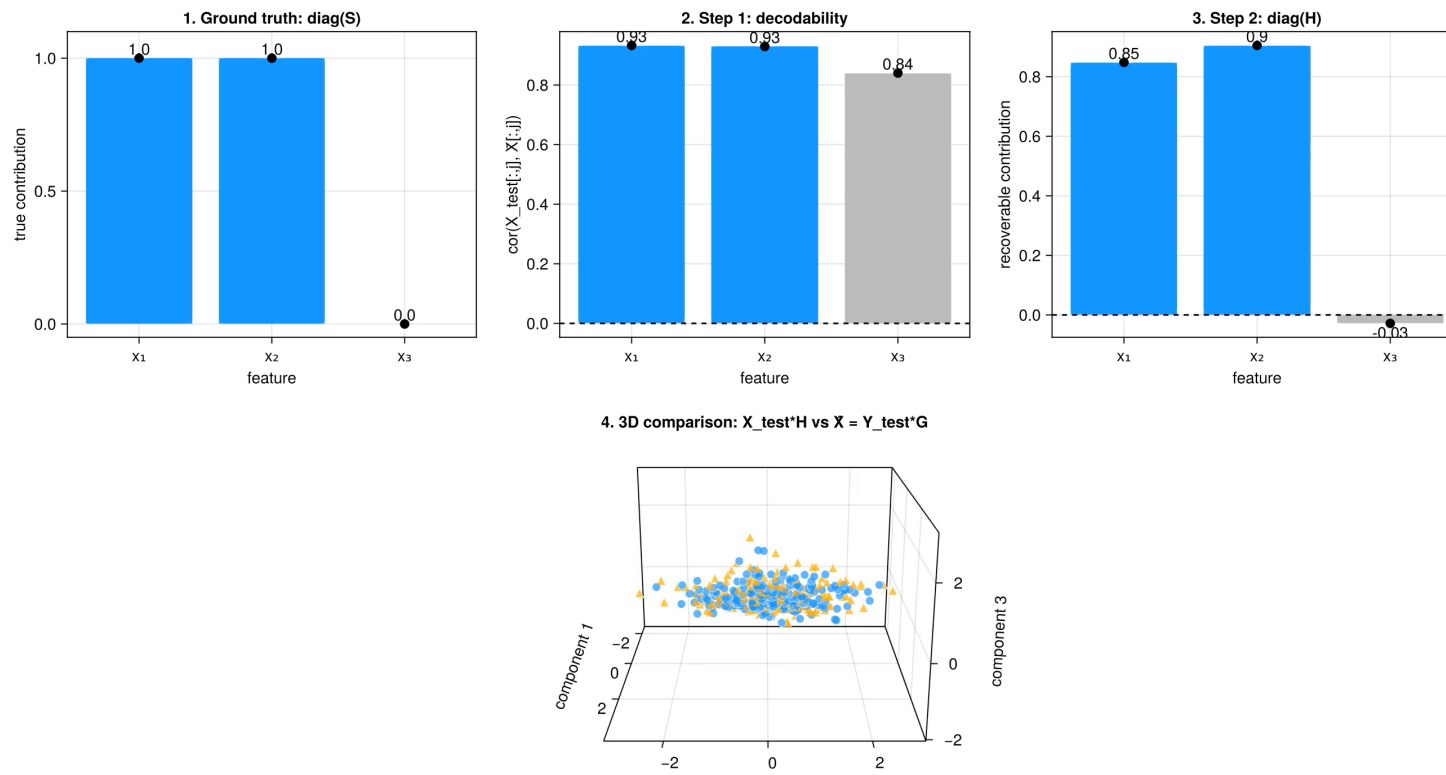
Output: estimate of prediction improvement $\Delta R \in \mathbb{D}^{d_x}$.

```

1  $H, G = \text{B2B}(X_{train}, Y_{train});$ 
2  $R_{full} = \text{corr}(X_{test}H, Y_{test}G);$ 
3 for  $i = 1, \dots, d_x$  do
4    $K = Id;$ 
5    $K[i] \leftarrow 0;$ 
6    $R_k = \text{corr}(X_{test}KH, Y_{test}G_i);$ 
7    $\Delta R_i = R_{full} - R_k;$ 
8 end
9 return  $\Delta R$ 
```

Demo:

- x_1 and x_2 are true contributing features; x_3 is correlated with x_1 but does not directly contribute to Y .
- Although x_3 is decodable due to its correlation with x_1 , B2B assigns it no independent contribution.



Top row: true contribution → what is decodable → what B2B recovers. Bottom: blue = $X_{\text{test}}*H$, orange = $X = Y_{\text{test}}*G$.

Zurich Cognitive Language Processing Corpus (ZuCo)

(Hollenstein et al., 2018).

ZuCo dataset provides co-registered eye-tracking and EEG data during natural reading:

- 12 subjects, native English speakers
- Reading natural English text 4-6 hours
- 3 tasks
- 21,629 words
- 1107 sentences
- 154,173 fixations
- 128-channel EEG acquisition

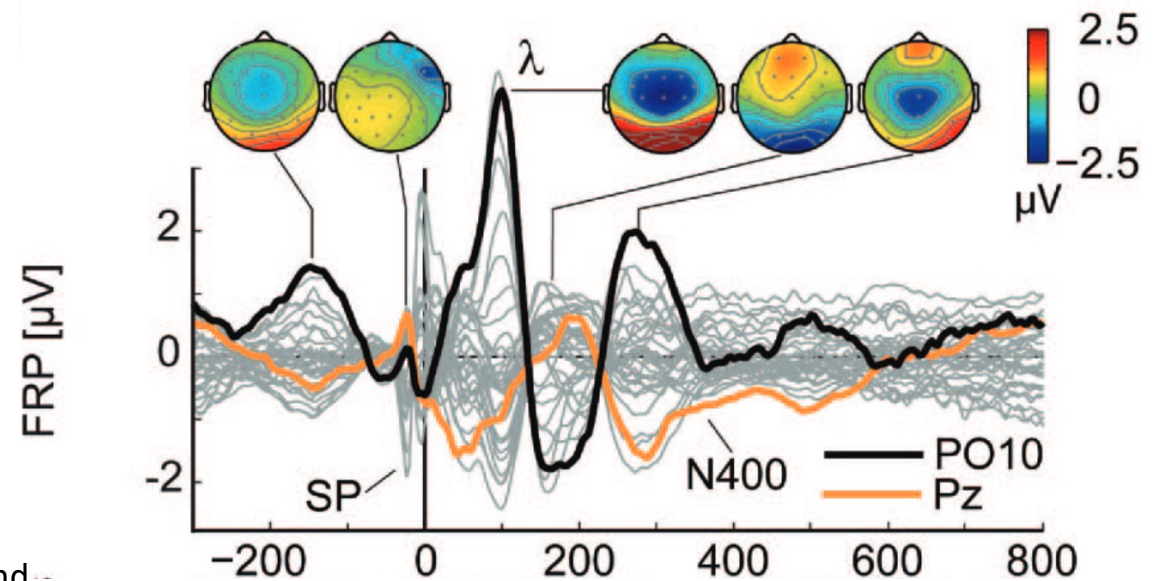
Is B2B sufficient?

Challenge: Temporal Overlap in EM-EEG co-registration:

- Multimodal data: neural responses + eye movements.
- Fixations occur in rapid succession during reading.
- EEG responses to adjacent fixations overlap in time.
- The observed EEG signal may reflect multiple events and predictors simultaneously.

Fixation-related Potentials (FRPs) & Overlaps

- Event-related EEG responses time-locked to the onset of eye fixations.
- They contain both early visual responses and later language-related components, such as the N400.
- Critically, these components can overlap with responses from neighbouring fixations.



Source from Dimigen et al. (2011)

Combine overlap correction and b2b regression

Research Questions:

1. In simulations mimicking co-registration data with known ground truth, does combining overlap correction (rERP) with B2B improve the disentanglement and recovery of collinear predictors under temporal overlap?
2. When applied to ZuCo, does the best-performing pipeline yield stable and interpretable estimates for lexical predictors (surprisal, frequency, length) while controlling for oculomotor covariates?

Simulation design:

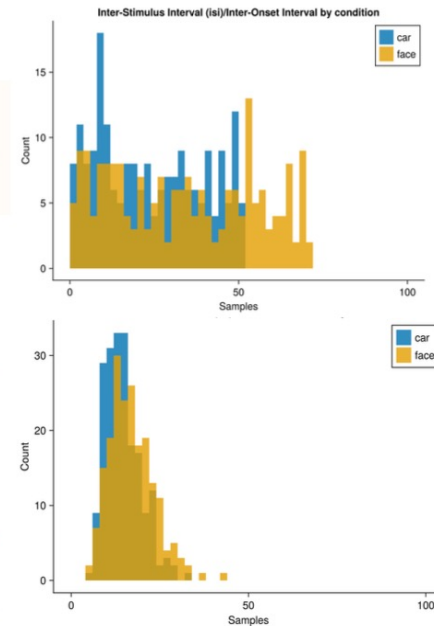
- Simulate multi-source pre waveform shapes with known
- Temporal overlap: none vs
- Predictor collinearity: non

```
# uniform onset formula
#o = UniformOnset(; width = 50, offset = 0);
o = UnfoldSim.UniformOnsetFormula(
  width_formula = @formula(0 ~ 1 + condition),
  width_beta = [50.0, 20.0], # longer width, less overlap for face condition
  # bias inter-onset intervals by condition to modify the overlap.
)
```

```
# logNormalOnset
#log_n.o = LogNormalOnset(; μ = 3, σ = 0.25, offset = 0, truncate_upper =
nothing);

target_mean_ms = 250.0
target_mean_samples = target_mean_ms/1000 * sfreq
σ = 0.35
μ0 = log(target_mean_samples) - (σ^2)/2

log_n.o = LogNormalOnsetFormula(
  μ_formula = @formula(0 ~ 1 + condition + continuous),
  μ_beta = [μ0, 0.10, 0.05], # [Intercept, condition(face), continuous]
  σ_beta = [σ],
  offset_beta = [0.0],
  truncate_upper = nothing,
)
```



Conceptual Methods Map

		Collinearity	
		Low	High
Overlap	None	Standard decoding <i>baseline</i>	Back-to-back regression (B2B) <i>disentangles collinear predictors</i>
	High	Overlap-corrected decoding <i>deconvolution rERP</i>	One-step: deconvolution rERP + B2B Two-step: GLM-reconstructed trials → B2B

Deconvolution rERP:

- Modelling continuous EEG with a time-expanded design matrix, allowing overlapping event-related responses to be estimated jointly, and provides overlap-corrected estimates.

GLM-reconstructed trials:

- Use the GLM estimates to reconstruct the activity overlapping with a target fixation, subtract this activity from the continuous EEG signal, and then epoch the target fixation.

Simulation evaluation:

Cross-talk matrix: $M_{ij} = \text{corr}(\hat{w}_i(t), w_j^{\text{true}}(t))$

- $\hat{w}_i(t)$ denotes the waveform recovered for predictor i .
- $w_j^{\text{true}}(t)$ denotes the true simulated waveform for predictor j .
- Ideally, the matrix should show high values on the diagonal and low values off the diagonal.
- Indicating that each true effect is recovered for the correct predictor with little leakage to other predictors.

Real-data application: *ZuCo*

- Select the best-performing pipeline(s) based on simulation performance, carry forward to *ZuCo* analysis

$$EEG(t) = \beta_0(t) + \beta_{\text{surprisal}}(t) \text{ surprisal} + \beta_{\text{freq}}(t) \log \text{freq} + \beta_{\text{len}}(t) \text{ length} + \beta_{\text{ocul}}(t) \text{ oculomotor} + \varepsilon(t)$$

Lexical factors:

- Word predictability (surprisal)
- Word frequency (log)
- Word length (z-score)

Oculomotor controls:

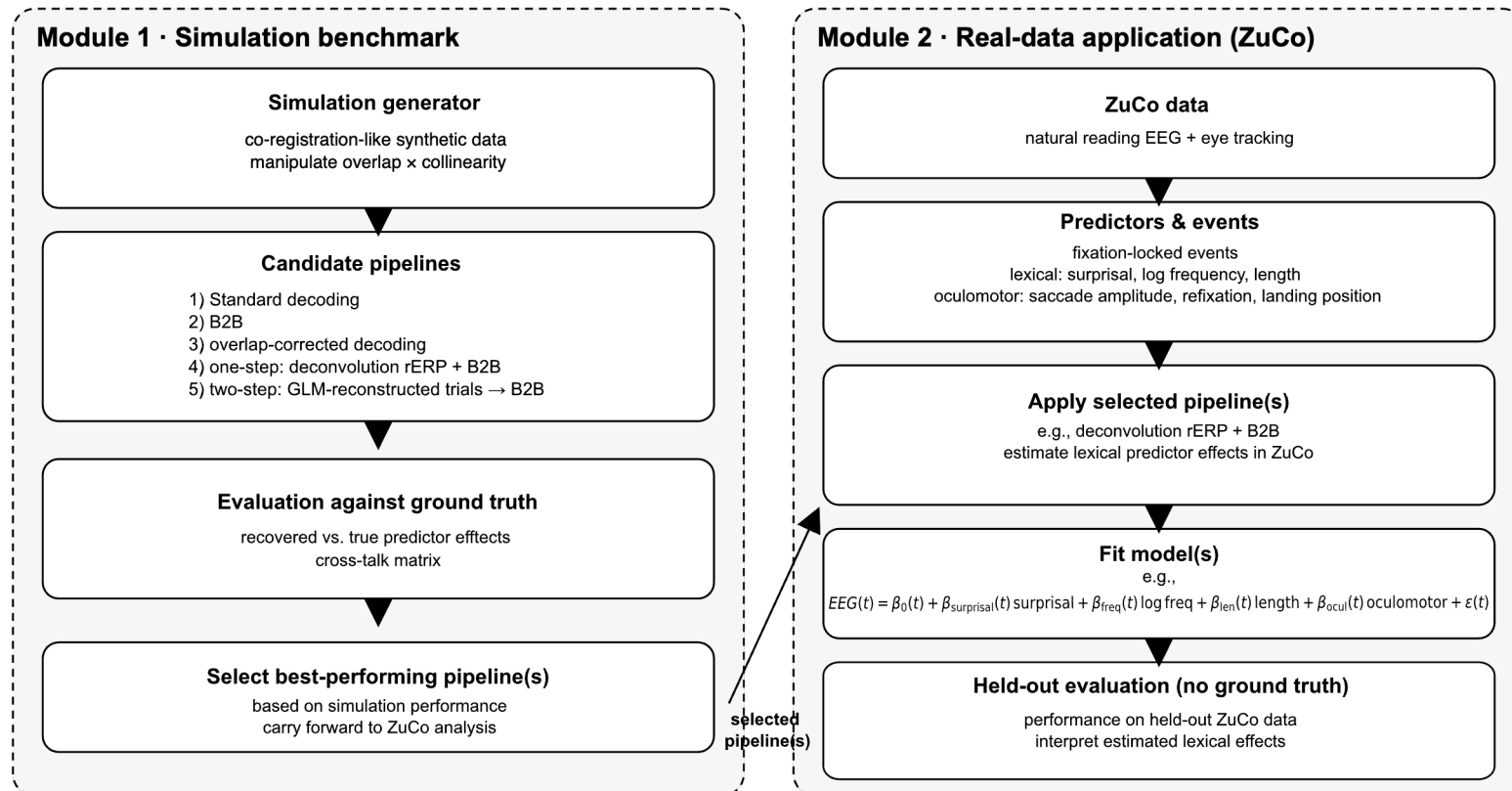
- Saccade amplitude
- Refixation
- Landing position

- ➡ • Word surprisal ($-\log p(w_i | w_{<i})$), derived from a pretrained language model as the negative log-probability of each word given its preceding context.
- Will be aligned to the corresponding fixation events in *ZuCo* and included in the model as a lexical predictor.
- A fixation-aware variant will additionally distinguish first fixations and refixations.

- ➡ Real-data evaluation:
 - Ground-truth neural effects are unknown.
 - Follows the feature-importance evaluation proposed in the original B2B framework.

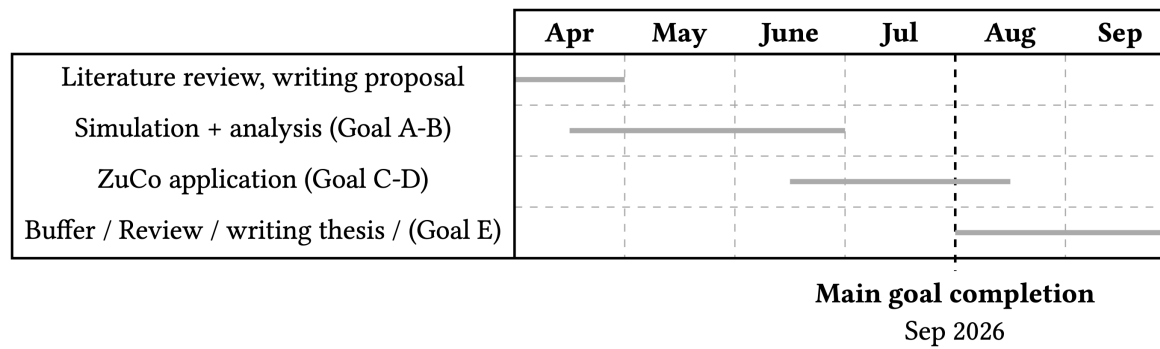
Project workflow overview:

Simulation benchmarking → select best pipelines → ZuCo application



Plan:

- Review the relevant literature on B2B, FRPs, and the deconvolution rERP.
- Implement a simulation benchmark with known ground truth, and evaluate alternative pipelines under controlled overlap and collinearity.
- Apply the best-performing configuration(s) to the ZuCo dataset.
- Result consolidation, documentation, and thesis writing.



References:

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